

The Suppliers of Components for Electric Lamps

359. We have referred briefly in paragraph 11 to the suppliers of the principal components for reference lamps, namely, Lamp Caps, Glass Bulbs, Lamp Metals and GTC, the four companies owned jointly by BLI (through AEI L & L) and GEC; Lamp Presscaps, a subsidiary of BLI; Mullard, a subsidiary of Philips; and Chance Bros, a subsidiary of Pilkington. At the time of our predecessors' inquiry in 1949-51, there were only two components companies jointly owned by AEI and GEC, Lamp Caps and Glass Bulbs. The position in 1951 and the considerations taken into account by our predecessors regarding the supply of components by these two subsidiaries of members of ELMA, and by the other members of ELMA, are outlined in paragraph 161. We have set out in paragraph 114 our predecessors' recommendation that ELMA should undertake that members who sold lamp components (other than patented components and ready-coiled filaments) would make them equally available to members and non-members at prices which should not be higher to non-members than to members, subject possibly to lower prices being charged to parent, subsidiary or fellow subsidiary companies of the supplier. ELMA's subsequent undertaking relating to this and our predecessors' other recommendations are dealt with in paragraphs 167 and 170.

360. *Lamp Caps Ltd.*, the first of the jointly owned components companies to be formed by AEI and GEC, was set up in 1922 to take over the then parent companies' cap making interests. Until 1967 Lamp Caps made a range of the common bayonet type, and also the screw type (still used generally on the Continent), of vitrited (glass insulated) caps for filament lamps and some discharge lamps; Lamp Presscaps (a subsidiary of Thorn until 1964) made a range of vitrited caps, and was the sole supplier in this country of bi-pin caps for fluorescent lamps. Lamp Caps and Lamp Presscaps, between them, supplied virtually the whole of the cap requirements of the electric lamps industry in the United Kingdom. They also exported caps. Relatively small quantities of lamp caps were, and are, imported; mainly automobile and other special purpose types from NV Vitrite, a subsidiary of NV Philips, which has no selling agent in this country.

361. We understand that by 1959 it had become apparent that cap manufacturing capacity in this country would need to be expanded, and there were preliminary discussions between Thorn and the parent companies of Lamp Caps about the possibility of the formation of a joint cap factory. There was no space for expansion at Lamp Presscaps' factory at Edmonton and efforts to increase production at the factory raised certain problems, including that of effluent treatment. We were told that Lamp Presscaps was faced with the alternatives of establishing a new factory almost certainly some distance away or of merging in whole or in part with Lamp Caps which had room to expand at its Chesterfield factory. Both possibilities were discussed both before and after the formation of BLI in 1964, but little progress was made. The minutes of a meeting of the Thorn Executive Board held on 14th August 1964 recorded that it had been agreed that sufficient time had elapsed since the formation of BLI to make it permissible for Lamp Caps and Lamp Presscaps to increase their prices to outside customers. In reference to this record, BLI has explained that before the announcement of the merger at

the end of May 1964 'both Lamp Caps and Lamp Presscaps had independently decided to increase cap prices due to greatly increased manufacturing costs, in particular, due to the price of copper. After the formation of BLI it became apparent that both companies had good reason to increase prices but it was considered that it would be unwise to announce these increases under the immediate impact of the announcement of the formation of BLI. The minute extracted meant that it was then considered that sufficient time had elapsed since the announcement of the merger for the previously planned price increase by Lamp Presscaps to be put into effect.' In August 1966 Lamp Caps told us that its selling prices to the parent companies were based on 'cost plus five per cent'. No price list or quantity discount scales had been published since 1957, but the potential value of annual business was said to be taken into account in quoting prices to other customers.

362. In order to illustrate the principles involved in its sales policy, Lamp Caps submitted a schedule showing its 1965/66 selling prices to a number of customers of the three types of vitrited cap (Standard Bayonet BC, Edison Screw E27 and Goliath Edison Screw E40) which, together, represented about 89 per cent of its total sales, by value, of caps for use in the manufacture of reference lamps in 1965/66. The following is an extract from the schedule:

Customer	Quantities ('000)			Price per 1,000		
	BC	E27	E40	BC	E27	E40
				£ s. d.	£ s. d.	£ s. d.
Osram	52,049	10,424	1,540	2 15 3	2 12 9	9 5 9
AEI L & L	39,461	2,052	1,323	2 15 3	2 12 9	9 5 9
Together	91,510	12,476	2,863	—	—	—
Britannia	11,553	200	40	3 6 3	2 19 3	11 16 9
Ismay	10,405	750	55	3 6 3	2 19 3	11 16 9
Evenlite	125	—	—	3 8 0	—	—
Crompton	16,780	1,500	—	3 0 9*	2 18 3	—
British Luma	240	—	—	3 6 6	—	—
Kingston	2,692	250	212	3 6 3	3 1 6	12 3 0
Philips Electrical†	3	301	111	3 13 3	3 2 3	13 1 3

*We were told that this was a special price for caps delivered in bulk containers and that this method of delivery represented a considerable cost saving to Lamp Caps.

†In 1965/66 Philips Electrical obtained the bulk of its requirements of standard vitrited caps from Lamp Presscaps.

363. Discussions about a joint cap factory were resumed in 1966 and it was finally agreed to concentrate production of vitrited caps at Chesterfield by the purchase by Lamp Caps of Lamp Presscaps' business in this type of cap, and to expand Lamp Presscaps' production of bi-pin caps at Edmonton. The merger became effective on 1st January 1967, and for the next four months Lamp Caps and Lamp Presscaps were involved in the transfer of machines and stocks and in interim arrangements for the maintenance of supplies. Before the merger Lamp Caps supplied about 70 per cent by number of the vitrited caps produced in this country and Lamp Presscaps supplied the other 30 per cent. Lamp Caps is now the sole manufacturer in the United Kingdom of vitrited caps; it continues to make special types of bi-pin caps for certain discharge lamps and has a very small trade in pressings for electronic components. Lamp Presscaps, whose present activities are dealt with in paragraphs 367 and 368, continues to be the sole manufacturer in the United Kingdom of bi-pin caps for fluorescent lamps. (The

company's separate manufacturing facilities for products not concerned with electric lamps, which were transferred to Thorn in 1964, remain under its management and the costs of management are recovered in full from Thorn.)

364. The terms on which Lamp Caps acquired the vitrited caps assets of Lamp Presscaps were not embodied in a formal agreement, but were set out in the following extract from the minutes of a meeting of the Board of Lamp Caps held on 23rd November 1967:

Integration

The following report on the Integration of Vitrited Cap Production was approved and adopted.

1. The whole of the plant formerly operated by Lamp Presscaps Limited for the manufacture of glass insulated caps has been purchased by the Company for the sum of £90,561, which represents the written down valuation for Corporation Tax purposes at the 31st March 1966, plus the cost of additions since that date, plus the agreed sum of £5,000. Spare parts, furnace blocks and tools (valued according to the basis of valuation adopted by the Company in its financial accounts) amounting to £30,480 have also been purchased.

2. The dismantling of the Lamp Presscaps Limited plant and its transport to Chesterfield has cost £6,502 and these costs have been paid by the Company as agreed.

3. All Lamp Presscaps Ltd. stock holdings relating to vitrited caps have been transferred to the Company and the valuation has been in accordance with the principles adopted by Lamp Caps Limited when valuing stocks. Stocks acquired were as follows:

(a) raw materials and components	£32,535
(b) work in progress	£58,149
(c) finished goods	£63,652

The valuation of each line of stock has not exceeded the relevant selling price less the selling costs and it has been on this basis that the provision for obsolescent and slow moving stocks has been computed.

4. The cost of maintaining stocks at the Lamp Presscaps Limited store since the agreed date of 31st December 1966 amounting to £2,009 has been met by the Company.

5. Lamp Presscaps Limited initially despatched some of the stock direct to customers, and in so doing incurred distribution costs of £330. The Company has reimbursed Lamp Presscaps Limited for this sum.

6. Revenue expenditure of a non-recurring nature, amounting to £25,067, incurred in relation to the acquisition of the plant and stocks from Lamp Presscaps Limited, has been carried forward in the accounts at 31st March 1967. This sum will be written off in the accounting period 1st April 1967 to 31st March 1968.

At the same Board meeting it was agreed that Lamp Caps would pay a service charge to BLI for its management of the company. Starting on 1st April 1967 for a period of seven years ending 31st March 1974, when the net profits of Lamp Caps (before corporation tax) exceed £100,000 in any one year, a service charge will be paid to BLI by Lamp Caps made up as follows:

- (a) the first additional £25,000 in excess of £100,000
- (b) 20 per cent of any further profits in excess of £125,000.

If the aggregate service charge received by BLI during the seven year period totals less than £250,000, supplementary service charges will be payable in the three years ending 31st March 1977. These supplementary charges will be limited to £25,000 in each year representing the first £25,000 profit earned by Lamp Caps in excess of £100,000 each year. The total of the supplementary service charges will be limited to the amount by which the service charges payable in respect of the seven years ending 31st March 1974 falls short of £250,000.

365. In December 1967 Lamp Caps clarified its sales policy as it had stated it to us in August 1966 before the merger (see paragraph 361). It said that although no price lists or quantity schedules had been published since 1957, that was not to say that at any time customers and the company's own sales staff were not aware of the prices at which the company was prepared to supply caps for reference lamps in normal demand. The prices were based upon the history of past trading and the normal commercial processes of quotation and negotiation, and they reflected the volume of the customer's business, the company's costs and the margin of profit aimed at and also the competition from foreign suppliers. For example, a general price increase of $1\frac{1}{2}$ per cent with an additional 10 per cent on nickel plating charges made in October 1967 was occasioned by the surcharge on fuel oil prices and an increase in the price of nickel.

366. Later in the same month Lamp Caps clarified the position further, and said that:

- (i) before 1951, it had three price lists; to parents, to other members of ELMA and to all other customers;
- (ii) after the undertaking in 1951 not to differentiate between customers, from 1951 to 1957 there were two price lists; one to parents, and one to all other customers based on standard prices and predetermined quantity discounts.

It was found that each of the price lists at (ii) had drawbacks. In the case of the parents, it meant that the parent taking the larger quantity was contributing to the other parent's dividends as the dividends were shared equally. In the case of other customers, the fixed prices and quantity discounts made it difficult to compete with Lamp Presscaps and NV Vitrite. In 1957 the then existing price lists and discounts were abandoned, and the following new arrangements were introduced:

- (i) parents took supplies at 'cost plus 5 per cent', which terms still apply;
- (ii) initial prices to all other customers were the same as the previous standard list prices less appropriate predetermined quantity discounts. These prices have been varied by negotiation, as described in paragraph 365.

367. *Lamp Presscaps Ltd.* continued to produce vitrited caps until June 1967; these were supplied to Lamp Caps at Lamp Caps 'factory transfer prices'. Lamp Presscaps also recovered from Lamp Caps certain expenses of stock maintenance and distribution as provided by the terms of the merger (see paragraph 364). We were informed that before the merger of cap interests Lamp Presscaps' prices to BLI were negotiated 'at arms length', and that all other customers paid 'the same sort of prices' quantity for quantity.

368. Further inquiries showed the following prices charged for the main types of caps in 1966/67.

	Prices per 1,000 caps					
	Vitrited caps				Bi-pin caps	
	Standard bayonet		Edison screw			
	s.	d.	s.	d.	s.	d.
BLI	60	3	57	9	72	0
Osram	—	—	—	—	85	0
Philips	71	9*	72	0	—	—
British Luma	74	0	—	—	—	—
Maxim	75	0	—	—	—	—
BELL	76	0	—	—	—	—

*The cap supplied to Philips at this price varies slightly from the standard type and is marginally more costly to produce.

The price to Osram for bi-pin caps was reduced after the merger to 75s. Od. per 1,000. Although this reduction was not a condition of the merger, it may be said to have resulted from it in that it was considered that, in the new conditions, the differential between the prices to BLI and to Osram should be less marked. From the details shown above and certain explanations given to us, it appears clear that prices charged to BLI were comparable with the standard price to parents charged by Lamp Caps, and that the prices to other customers were related roughly to quantities. We understand that prices of standard bayonet caps to Philips were fixed at a level to obtain the best price without encouraging Philips to import from NV Vitrite. (Bayonet caps would be a low volume production item for NV Vitrite, and consequently relatively costly.)

The observations of other manufacturers on the Lamp Caps/Lamp Presscaps merger

369. Following the publication of the proposed acquisition by Lamp Caps of the interests of Lamp Presscaps in vitrited caps, the Board of Trade announced that they would not refer the merger to this Commission. The Board told BLI and GEC that in reaching this decision they had had in mind the fact that the supply of certain descriptions of electric lamps was currently under investigation by the Commission, and that no doubt the Commission would bear in mind the new situation created by the merger in their investigation of the lamps industry generally.

370. In furtherance of our consideration of this matter, in May 1967 we invited the manufacturers of reference lamps (apart from BLI and Osram whose separate comments on the merger, and also on the comments of the other manufacturers set out below, are in chapter 11) to submit any representations or comments on the merger which they might wish us to take into account.

371. Philips said on 14th July 1967

The recent acquisition by Lamp Caps Ltd. of Lamp Presscaps Ltd.'s business in vitrited caps has resulted, so far as we know, in a situation where Lamp Caps is the sole manufacturing source in the United Kingdom of vitrited caps. This cannot be said to be an ideal situation from the point of view of a lamp manufacturer dependent upon a supply of caps. Initially the Lamp Caps merger resulted in a deficiency in supplies to Philips Electrical both from a quality and quantity point of view and in consequence in the early months of this year we had to import by air, at very short notice, a quantity of lamp caps from Holland to bridge the gap caused by initial difficulties at Lamp Caps' Chesterfield factory. These initial difficulties however were soon overcome, and we wish to confirm therefore that the current position in relation to the supply of lamp caps from United Kingdom sources to meet our present requirements is reasonably satisfactory. In as much as we wish from time to time to import lamp caps from the continent, we would take this opportunity of pointing out that the burden of import duties on lamp caps is a matter which we cannot ignore. We somewhat doubt whether the supply position as it has now evolved in the United Kingdom really justifies in the national interest, the continuation of import duties on lamp caps. We have to date taken no steps to apply to the Board of Trade or Treasury for exemption from, or repeal of, import duties applicable to lamp caps. We would express the view however, that in the event of the supply of lamp caps from United Kingdom sources deteriorating, either as to price, delivery, quality or quantity, resulting in a need to increase our importation of lamp caps, we may well wish to refer the position to the Board of Trade for their consideration. We are watching the position carefully, and although at present the main GLS types of cap from Lamp Caps Ltd. are competitive on price, we will hope to see a continuing reduction in price on their supplies on certain types, particularly the smaller sign lamp caps. As to the availability of types from Lamp Caps Ltd. we have one problem in particular which

relates to caps for fluorescent tubes. The lamp caps manufactured by Lamp Caps Ltd. for fluorescent tubes are of a particular type of construction and appearance from which they can be identified as being Thorn caps. This presents certain marketing problems both in home and export markets. On the home market it results in a situation where the caps on Philips fluorescent tubes are indistinguishable from those on Thorn tubes. The problem is of greater significance in relation to exports because due to standardisation requirements of Philips on the continent, the Thorn type cap cannot be used on that part of our Hamilton production which is destined for export, and in this case we are obliged to import caps for export requirements.

372. *Luxram* considered that it was always better to have two firms rather than one negotiating for its business and added that the two cap companies 'have always kept their prices just about at the same level as we could buy from abroad, taking into consideration the duty which would be payable on imported material.' *Luxram* is buying caps independently of Philips. *Kingston* said 'We are now deprived of an alternative source of supply should one supplier be out of stock of any particular item; this merger took place only a short time ago and it is rather early to have any experience on which to comment properly.' *Kingston* is now buying against Philips' contract with Lamp Caps.

373. *Crompton* considered 'that a monopolistic situation exists particularly in relation to bayonet caps where the continent of Europe offers no competition. Substantial price increases have been made in the last few years and the latest price of the standard BC cap is the same as that for last year's contract despite a very considerable fall in the price of copper. As the only shareholders in Lamp Caps Ltd. are BLI and GEC it is in their interests but not ours to keep cap prices at a high level.' *Crompton* also considered that the prices charged by the two cap companies were unduly high compared to prices quoted by NV Vitrite because where there had been competition from NV Vitrite its prices, which include import duty, were often lower than those of the United Kingdom companies.

374. *BELL, Maxim and Viking* viewed the merger with suspicion, as it left only one source of supply of vitrited caps in the United Kingdom. *Maxim* said that now that all vitrited caps come from one supplier 'this obviously is a disadvantage unless of course Lamp Caps have given the Government an undertaking that supplies will be made freely available to the trade and that prices charged will be competitive with world supplies of caps.'

375. *Longlamps* said 'We were first advised of this amalgamation by Lamp Presscaps Ltd. on 12th January 1967 and whilst we have noted no change in the degree of service or availability, we have received from each company notification of price increases of the products of both and both have offered the reason for the increases as being due to the increased costs of raw materials, notably copper and brass. Both notifications of these price changes were received in March of this year.'

376. *Zolta and British Luma* considered that in the past they could buy vitrited caps from Lamp Presscaps at lower prices than were charged by Lamp Caps since the merger. *Zolta* also said that it had obtained quicker delivery from Lamp Presscaps. Details supplied by *British Luma* showed that since October 1965 the two cap companies had submitted level tenders for vitrited caps, whereas before 1965 Lamp Presscaps' prices had been lower.

377. *Hivac and Lymelite* had no comments on the merger.

378. *Glass Bulbs Ltd.*, the second of the jointly owned components companies to be formed by AEI and GEC, was set up in 1948 primarily to operate the two ribbon bulb making machines imported from the USA. It was estimated in 1951 that the potential annual output of all articles by the machines (which can make other products as well as bulbs for electric lamps) was 450 million. This potential, with such other machines as were retained for the production of special types of bulbs, was far in excess of the then requirements (and of the current requirements) of the United Kingdom lamps industry, and a large proportion of production was and is exported (65 per cent in 1967). In 1951 Glass Bulbs confirmed that the bulbs made on the ribbon machines would be distributed without discrimination, and that the prices charged would be the same—quantity for quantity—to all purchasers in the United Kingdom, with the exception that the proprietors would purchase at lower prices, as would the Controlled Companies since it was considered that the joint proprietors had an obligation to the other shareholders in the Companies. An agreement dated 14th December 1948 between BTH (subsequently substituted by AEI L & L), GEC and Glass Bulbs, which has been registered with the Registrar of Restrictive Trading Agreements, covered the grant by the shareholders to Glass Bulbs of patent licences and the provision of know-how, and it provided, inter alia, that the prices to be charged by Glass Bulbs to the shareholders should be total cost plus 6 per cent, or such other percentage as might be agreed by the shareholders.

379. Glass Bulbs publishes standard list prices for lamp bulbs. There are special arrangements for the parent companies, Philips and the Controlled Companies. Other customers buy at list prices less quantity rebates. These arrangements are said to implement the undertaking given in 1951. BLI and Osram confirmed our understanding that as bulbs are a standard product manufactured in long runs on highly automatic machinery the actual cost of manufacturing a bulb is the same for whichever customer it is produced. They said that the differences in the prices charged to different customers are due to commercial considerations which are based primarily on the buying power of a particular customer, but must also reflect the differences in cost of distribution.

380. Details of the pricing arrangements are as follows:

- (i) to parents, cost plus 6 per cent;
- (ii) to Philips, cost plus 17½ per cent;
- (iii) to the Controlled Companies, list prices less rebate of 17½ per cent;
- (iv) to other customers, list prices less the following quantity rebates.

Quantity (millions)	Rebate (%)
0 up to 2.5	nil
2.5 " " 5.0	2.5
5 " " 10	5
10 " " 15	7.5
15 " " 20	10
20 " " 30	12.5
30 and over	17.5

- (v) all customers receive a transit breakage allowance of 2½ per cent and, apart from parents and Philips, a monthly settlement discount of 2½ per cent.

Glass Bulbs explained that the terms to Philips were related to a 10-year contract covering total annual purchases in excess of 30 million bulbs. The contract expired in 1966 but we understand the terms are unchanged, as is the minimum qualifying annual quantity taken by Philips.

381. The above pricing arrangements resulted in the following actual prices paid by a selection of customers for two types of bulbs required in large volume:

Type	Standard list price	Price per 1,000 bulbs				
		Parents 1965 Cost +6%	Phillips 1966 Cost +17½%	Controlled Companies 1965/6 List -17½%	Crompton 1965/6 List -12½%	Br. Luma 1965/6 List -5%
		<i>s. d.</i>	<i>s. d.</i>	<i>s. d.</i>	<i>s. d.</i>	<i>s. d.</i>
R/PS60 pear shaped } clear		75 0	41 2	61 11	65 7	71 3
25-100 watt } pearl		105 0	52 3	86 8	91 10	99 9
R/M55 mushroom } clear		80 0	40 11	66 0	70 0	76 0
60 watt } pearl		107 0	53 10	88 3	93 7	101 8

Notes: (i) The pear shaped bulb (R/PS60 pearl and clear) accounts for nearly half total bulb production.

(ii) Prices exclude transit breakage allowances and settlement discounts to Britannia, Crompton and British Luma.

The above prices paid by parents and by Phillips were equivalent to standard list prices less the following rebates:

	Parents List less (%)	Phillips List less (%)
R/PS 60 clear	48.7	45.1
pearl	53.5	50.2
R/M 55 clear	50.6	48.9
pearl	52.5	49.7

382. Glass Tubes and Components Ltd. (GTC). The formation of Glass Bulbs had left AEI with some glass production at Chesterfield and GEC with glass works at Wembley and Lemington-on-Tyne; both companies also had lamp metals operations at five locations remote from their respective main lamp factories. In 1959 AEI and GEC began negotiations for the integration into two additional jointly owned companies of such of their lamp components operations as were not already combined in Lamp Caps and Glass Bulbs. The negotiations resulted in the formation in 1961 of GTC, which has factories at Chesterfield and Lemington-on-Tyne, and of Lamp Metals, which has factories at Wembley and elsewhere. At Chesterfield GTC makes soda glass tubing for fluorescent lamps, lead glass tubing for flanges (or flares) including some finished flanges* and rod and exhausts for filament lamps. It also makes domestic glassware on machines which were used for lamp bulb production before the setting up of Glass Bulbs. The Lemington factory produces special bulbs, including those for mercury and sodium lamps, projector lamps, lighthouse lamps etc; it also makes glass products not connected with electric lamps.

383. Prices to the parent companies are at full cost. A standard cost is prepared for each item and deliveries to parents are invoiced on this basis; the prices are

*Most lamp manufacturers, including BLI and Osram, make their own flanges from glass rod; some, including Osram, also supply them. GTC, which draws part of its sales requirements of flanges from Osram, keeps three flange machines in use partly as a buffer against any shortages experienced by the parents, and partly to supply some of the requirements of the Controlled Companies and to supply the requirements of Maxim.

adjusted annually and corrections are back-dated. Standard price schedules with, for certain items only, quantity discounts, are drawn up for bulbs, glass tubes for fluorescent lamps and glass tubing and rod; these schedules are not published but are said to be 'available to all customers who make application for them'. GTC said that a wide variety of specifications and types of finishes exist, and the schedules provide only a general guide to prices. Costs are considerably affected by the particular tolerances required by individual customers.

384. Examples of the prices charged to parents and others in 1965/66 are as follows:

(a) Soda glass lamp bulbs

	Invoiced price to parents £ s. d.	Price per 1,000 bulbs List price to others £ s. d.	
		£	s. d.
R.150A high wattage filament	17 14 3	44	11 6
KF46A twisted candle	8 2 0	16	2 6
X75A 75 mm discharge	10 4 6	14	10 6
T51BB discharge	17 17 6	24	0 0

(b) Lead tubing

Standard prices

Flare	9-15 mm	21.6 pence per lb
Exhaust	3- 5 mm	27.0 pence per lb
Rod	3- 5 mm	27.4 pence per lb

Prices are reviewed annually and discounts on tonnages above 20 tons are

20-99 tons standard price less approx 10 per cent

100-199 tons standard price less approx 12½ per cent

Actual prices charged in 1966:

	Invoiced prices to parents d.	Pence per lb		
		Philips d.	Controlled Companies d.	British Luma d.
Flare				
9-12.75 mm, diameter				
0.7-1.4 mm wall standard length	12.44	20.8	17.8	20.8
		(special tolerance 0.50 mm)		
Exhaust				
3-4 mm, diameter 0.4-1.0 mm wall 90-120 mm length	24.05	—	36.7	36.7
		(special tolerance 0.25 mm)		
Rod				
3-4 mm diameter 20-29 mm length	26.00	41.1	41.1	41.1
		(special tolerance 0.25 mm)		

(c) Fluorescent tubing

<i>Standard prices</i>	End trimmed Osram d.	Pence per tube	
		Shoulder formed BLI d.	Others d.
4 ft 1½ in diameter	4.0	4.75	5.75
5 ft 1½ in diameter	5.0	6.00	7.9

Actual prices charged in 1966 for shoulder formed:

	Invoiced prices to BLI d.	Pence per tube	
		Prices to Viking d.	
4 ft length	4.75	5.75	
5 ft length	6.00	7.90	

Sales to Viking, the only substantial outside customer for tubes, are 4.1 per cent by volume of total sales of tubes. Viking does not qualify for the quantity discount of 5 per cent on annual purchases of 1 million tubes.

(d) Flanges

	1966/67				Price per 1,000 flanges 1968			
	Parents <i>s. d.</i>		All others <i>s. d.</i>		Parents <i>s. d.</i>		All others <i>s. d.</i>	
35.0 mm diam } 21.0 mm length }	18	10	26	0	19	7	26	0
38.0 mm diam } 21.0 mm length }	20	2	23	6	21	8	23	6

(The above are the only types of flanges made by GTC)

385. *Lamp Metals Ltd.* has factories at Wembley, Rugby, Corby, Ponders End and Shaw and has recently established another factory at Gateshead. The factories at Corby, Ponders End and Shaw have now been closed. Processing of the raw materials is carried out at Rugby and manufacture of wire and other components is carried out at Wembley and Gateshead. (The company does not supply coiled filaments.) About three-quarters, by value, of total sales is made to the parents, and most of the remainder is sold to the Controlled Companies and BELL.

386. The company told us that before February 1965 prices to the parents were fixed at a level which would recover total costs and that adjustments were made at the end of each accounting period. Since then, prices had been related to the prices at which the parents could purchase the items from other suppliers. In practice, the policy has resulted in the prices to parents being identical with Mullard's prices, and any losses arising therefrom have been borne by the parents. For the small sales to other customers, each inquiry is negotiated in relation to the prices and quantity discounts of other suppliers (Mullard).

387. *Costs and profits.* The trading results of the four jointly owned companies are dealt with in chapter 10.

388. *Mullard Ltd.* makes tungsten wire, molybdenum wire, copper clad wire, coiled filaments and electrodes for reference lamps at its factory at Blackburn for supply to Philips Electrical and to all the other United Kingdom lamp manufacturers. Until recently, it also supplied glass tubing and rod to Philips Electrical and other companies in the group. Mullard has said that in its supply of components for reference lamps, it does not discriminate between its customers. Prices are on a quantity discount basis with the exception of associated companies which are at an inter-company price based on manufacturing cost. Standard list prices are published for tungsten wire, molybdenum wire and copper clad wire. There is no standard price list for coiled filaments. Mullard explained that filaments, even for GLS lamps made to BS 161, are by no means standard products; the specification allows manufacturers a choice of compromises and they all call for slightly different specifications of filament performance. However, the possible variations in filament dimensions are, in practice, minimal. Electrodes are manufactured to individual customers' own specifications. The glass tubing and rod, which Mullard has now ceased to supply, was sold to Philips Electrical at inter-company prices and to Luxram at prices competitive with those offered by other suppliers.

389. Prices to individual customers are determined on the following bases:

tungsten wire

- 1 m.-5 m. metres per annum—list less 10 per cent
- 5 m.-20 m. metres per annum—list less 20 per cent
- Over 20 m. metres per annum—list less 25 per cent

Exceptionally, BLI, which takes over 100 m. metres per annum, buys at specially negotiated net prices; and Crompton, whose arrangements are governed by a supply agreement dating back to 1957 when Crompton closed down its own wire production, also buys at special prices.

molybdenum wire

This wire is used only in small quantities in the lamp industry. Quantity discounts of up to 20 per cent are offered, according to quantity and specification.

copper clad wire

The standard list prices are net to all customers.

coiled filaments

Individual quotations have established a price pattern which varies only within 10 per cent over the range of customers.

electrodes

The individual specifications required preclude any standard prices or valid comparisons of prices between customers.

390. *Chance Bros. Ltd.* manufactures soda lime tubing for fluorescent lamps and lead glass tubing for flanges and exhausts at its factory at Smethwick. It does not supply finished flanges. Chance has no published price lists.

391. Pricing arrangements are as follows:

Soda lime tubing

Chance has for internal use standard price lists for the different specifications and lengths of tubing, and a scale of quantity discounts based on quantities from 250,000 to 3 m. The smaller manufacturers are supplied on this scale. Special arrangements are:

BLI takes the bulk of its supplies in end-formed tubes and has been on a special quantity scale of an additional 1 per cent per m. up to 16 m., then by $\frac{1}{2}$ per cent per m. up to 19 m. In 1965/66 BLI took quantities considerably above those envisaged by the special quantity scale and were given an additional rebate of £9,000. Subsequent to April 1968 the price list to BLI has been replaced by a new one which starts at 15 m. at approximately 5 per cent below the old 19 m. prices.

Philips takes all its supplies in trimmed and glazed tubes and until 1966 was buying from Chance at the top of the discount scale applicable to the standard price list. Chance then lowered its prices to meet the competition of GTC, but later increased them again and allows Philips an additional discount for increased quantity of 2 per cent for each extra million. As from 1st January 1968 a firm annual contract was negotiated at prices with an additional 17 per cent for the 4-ft tube and 12 per cent for the 5-ft tube.

Crompton and Viking. In 1967 Chance entered into a three-year contract to supply Crompton and Viking at special net prices on the basis of a joint require-

ment. It had formerly supplied each company at standard prices less the appropriate quantity discount.

Lead glass tubing and rod

Here again, Chance has standard list prices for the usual specifications of tubing and a quantity discount scale for internal use. The prices vary slightly according to the tolerance required, but Chance said that manufacturing efficiencies are now such as to make it possible to supply the tightest tolerances at no extra cost. The smaller manufacturers are quoted at standard prices and discounts. Special arrangements are:

Philips is supplied at a discount related to its potential purchases in order to encourage it to take larger quantities.

Crompton and Viking. The arrangements for soda lime tubing apply also to lead glass tubing and rod.